Final Project Design Phase

Computer Science - Week 11

Oct 3, 2026 - Version 0.0.4

For complete instructions and project ideas, refer to the assignment in Canvas. Specifically, review the final project requirements before you continue below. If you have not reviewed the project requirements in Canvas (you might have to go to the modules view and check the current or past week's content), you will not be able to properly complete the parts below. In addition, it may be helpful to review the information on this Drafter documentation page.

0) Project Planning and Preparation

Successful programmers spend time planning before they begin coding. *After* reviewing the project requirements, answer the following questions to help guide your design and development process.

A. What concepts from class or previous assignments might help you complete this project? Which programming concepts (loops, conditionals, lists, classes, etc.) do you expect to use? What programming skills do you want to improve while completing this project?

I will use dataclasses to store the State items as well as dataclasses to organize exercises. Also, I'll put each exercise user entry into a list so that all the data is cleanly organized. Then, I'll use matplotlib to allow the user to see a plot of their progression in an exercise. I think I'll be able to use conditionals to group exercises into a specific group based on the muscle group that they work. Also, to be able to plot exercise data, I'll need to use a for loop to go through a list of a particular exercise and multiply # of reps and # of sets and put that data into a new list so its graphable. While completing this project, I want to improve my debugging skills and organization skills while programming a project. I also want to create efficient code that is readable by other programmers. I want to improve my drafter skills in general and figure out how it actually works because that's still unclear to me.

B. Which parts of the project do you expect to be the easiest for you? The hardest? Why?

The parts that will be easiest are creating for loops to organize data from different exercise inputs since I got comfortable with using for loops over the past few weeks while studying for exams. Also, I think creating helper functions will be easy since I

have the most experience with doing that so far in this course. I also think using AI to assist me in the coding phase will make the entire process much easier. The hardest parts of the project will definitely be using drafter and route functions to bring together everything in the project. Also, creating the UI will be difficult since I've never done that before except some small parts in the homework involving textboxes and buttons. I think that after organizing all the data, the plotting part of the program shouldn't be too difficult since matplotlib is fairly simple to use.

C. Even though you have not yet designed the final project, estimate which tasks will likely require the most time and effort? Which features or components should be implemented first?

I think that creating the dataclasses (State and others) and route functions will take the longest time and the most effort because I've previously struggled with these on past assignments. I also think combining all the functions, route functions, and State to deliver a clean page with readable UI will take a lot of effort since this will be my first time doing something like this. The most experience I currently have is writing numerous functions like something from Project 1, but now I'll have to also write everything else to create a drafter Page displaying a program. In addition I think my project idea opens room for many edge cases especially with how many functions and components I'll have, so getting around that will take some time as well (writing unit tests). I'll probably start by writing my dataclasses first and building an outline of the 7 route functions and 5 pages required for the project. From there I'll write helper functions and then create route functions that will be called whenever a user presses a button or inputs something.

D. What distractions might interfere with your work and how will you minimize them? What resources will you need available while working? What questions, if any, do you still have about the assignment requirements?

I think my only distractions will be work from other classes and also the upcoming exam for this class. Apart from that I don't think any other distractions will affect my ability to interfere with this project. I can't really minimize those distractions but to help myself, I can split up the work for this project so that I don't try to cram everything in at once. While doing this project, an LLM will be very useful for me to troubleshoot issues, figure out solutions to a function, and also write code that I don't know how to write. I don't have any questions about the project requirements.

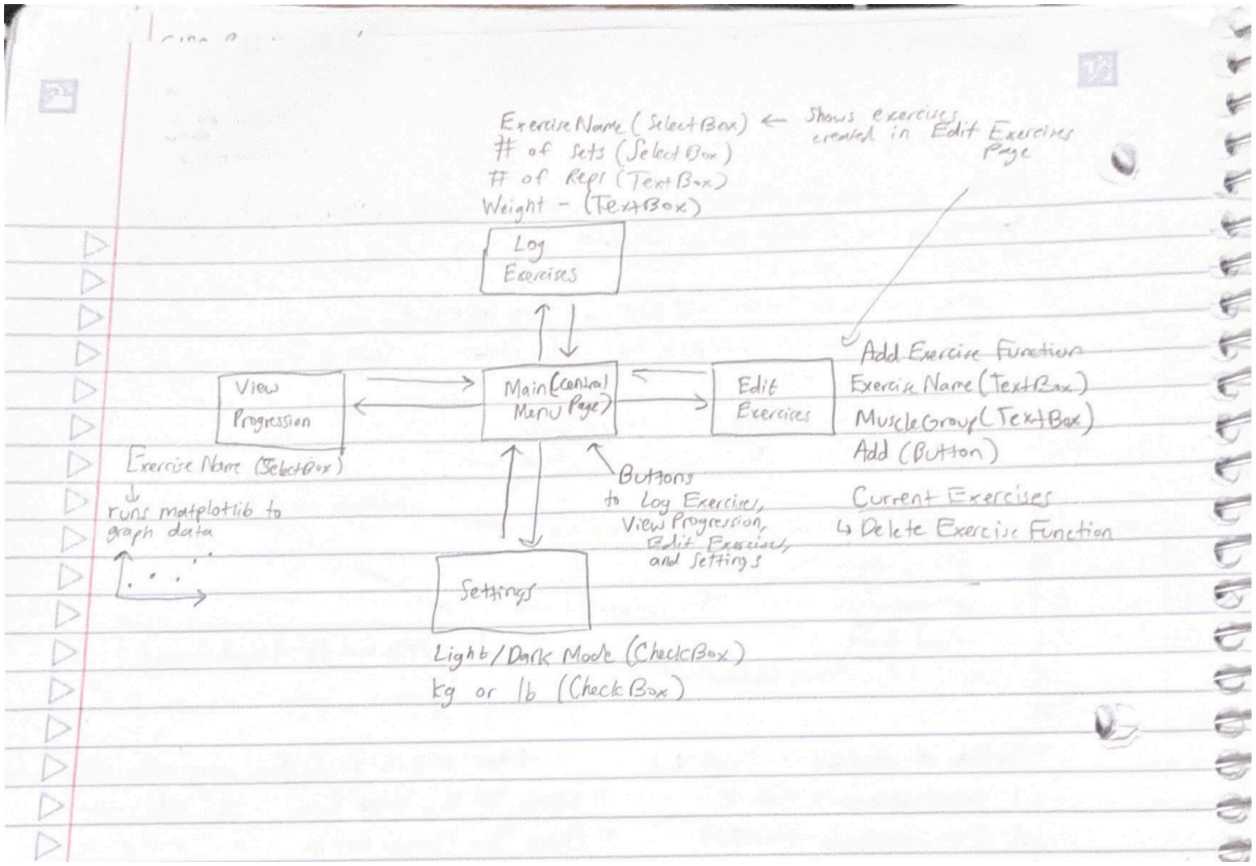
1) Site Diagram

Create a mock-up of your planned website. We suggest you make the mockup on paper, although you can also use software like PowerPoint or Slides to make a mockup. There is other design software that could also be used, but we strongly encourage you to keep it simple. Do not spend more than 15-30 minutes detailing the mock-up. Your goal should be to make a very quick and dirty diagram, to show that you have some kind of idea.

We recommend drawing each page as a rectangle, and laying out the text, buttons, and other components inside. Then, use arrows to indicate links between pages. Label each page the way you will name each function, and then put labels on the arrows to indicate what the button links. Also write down any parameters that should come into the pages.

In addition to the pages and routes, write down what fields your `State` will have, both their names and types. Think clearly about what data will need to be persistent across the entire website.

Add photo(s) or screenshot(s) of your website mock-up and labeled diagram on the next page.


Exercise Name (SelectBox) ← Shows exercises created in Edit Exercises Page
of Sets (SelectBox)
of Reps (TextBox)
Weight - (TextBox)
Log Exercises
View Progression
Main (Central Menu Page)
Edit Exercises
Add Exercise Function
Exercise Name (TextBox)
MuscleGroup (TextBox)
Add (Button)
Current Exercises
↳ Delete Exercise Function
Exercise Name (SelectBox)
↓
runs matplotlib to graph data
Buttons to Log Exercise, View Progression, Edit Exercises, and Settings
Settings
Light/Dark Mode (CheckBox)
kg or lb (CheckBox)

2) Implementing Requirements

This table should serve as a draft for how you will implement the requirements into your final project. You may change your ideas later on, but your final site must fulfill all the requirements.

Requirement	**Implementation**
Purpose/Functionality A 1-2 sentence description of the purpose or functionality of your site	**My site will allow users to track their workouts by inputting a workout, number of sets, and number of reps they were able to achieve. Then, they can view their progression for each exercise by viewing a plot.**
State Dataclass Attributes Minimum of 4 attributes, one of which must be a list/dict of instances of your second dataclass. Briefly describe each attribute, its type, and how it will be meaningfully modified throughout the site.	Exercises: list[Exercise] -> stores each exercise created by the user, will be modified whenever a user creates or deletes an exercise History: list[int] -> keeps a record of the weights for each exercise over time so the progression can easily be graphed. Will be modified as the user progresses in weights or even when their weight stays constant Streak: int -> keeps a record of how many workouts the user has done, will be modified each time a user logs a new workout session Filter: str -> filters through muscle groups like “Chest”, “Arms”, “Back”, “Legs”, will be modified whenever the user chooses a different muscle group filter
Second Dataclass Your second dataclass, including its attributes, their types, and how they will be meaningfully modified (if they are modified)	Exercise Name: str -> won’t change because the name of an exercise will remain the same each time Muscle_group: str -> won’t be modified, similar to Name Sets: int -> will be modified for each exercise but will

	largely remain the same for each user since usually people stick to a constant number of sets. Reps: int -> will be modified as a user gets more reps of a certain exercise over time. Weight: int -> will be modified rarely and over longer periods of time as a user slowly progresses in a certain exercise.
Routes (minimum 7) The names of your route functions	Index View_exercises Add_exercise Remove_exercise Progress_graph Clear_graph Settings_page
Pages (minimum 5) A brief description of the pages in your site	Main Menu -> this is where the user will start and see buttons that redirect to the other pages. Edit Exercises -> in this page, the user can add or delete exercises and assign them to a muscle group. These are the exercises that they can track and upload data to as they workout. Log Exercise -> this is where the user can select an exercise by name, log the number of sets they performed, the weight they did, and also the number of reps per set. View Progression -> this is where the user can view a graph for a given exercise that shows their overall progression using matplotlib Settings -> here, the user can change settings and small preferences in the program, like switching between dark/light mode and more.
User Input (minimum 3) Description of user input components (TextBox, TextArea, CheckBox, or SelectBox) and	Weight Lifted (TextBox) -> this is where the user enters in a weight (int) that they did in a particular exercise Sets Performed (SelectBox) -> This is where the user can select a number of sets that they performed

their purposes For more information on these input types, see Drafter Documentation.	(usually 1-4). Muscle Group (SelectBox) -> this is where the user can select which muscle group a particular exercise is hitting when they're adding exercises from a list of strings. Website Theme (CheckBox) -> can change between light and dark mode
If Statements (minimum 3) Brief description of what will be evaluated in your if statements and how they will meaningfully change the logic of the site	1. My first if statement will check if the textbox for exercise name is blank when a user creates a new exercise or edits a previous one. If blank, it will display an error telling the user to enter a valid name. 2. Max weight -> this checks if your current logged weight is greater than your highest ever logged weight and displays that as your max weight when you select that exercise. This sort of acts as a PR check. 3. Graphability check -> this if statement will check whether or not there's enough data points for a given exercise before graphing it with matplotlib. This is so that the user isn't able to view a graph with only one point, as that wouldn't show a progression.
Loop (minimum 1) This loop must iterate over the list/dict of dataclass instances in your State attribute. Description of this loop and why it is meaningful to the site	My loop will iterate over the history list in the State dataclass to grab weights for one particular exercise so that the data can be graphed into a plot showing exercise data for only one exercise. This will be meaningful since this feature (plotting progression) will be a good way for me to apply skills learned in this class by using matplotlib in a practical way to provide an informational graph for a user.

LLM Conversation

If you used an LLM in any part of this work, copy and paste every part of that conversation below. Otherwise, you may be given a zero for the assignment. Obviously, these words will not all be yours, and this is the exception to the requirement at the top of this assignment that all responses be in your own words. If you did not use an LLM to assist you in this assignment, simply say so below.

Did not use AI.

Final Submission

When you are finished, download this file as a Word Document (docx) and upload the file to the appropriate assignment on Canvas.

I'm working on a CS project that requires me to build a Python drafter program. I want to create a workout tracker that allows a user to log workouts (sets, reps each set (according to number of sets, weight) and also allows a user to modify existing workout names and then view progression across a workout exercise by using matplotlib. Please write me a program using the following requirements and technical direction that I've brainstormed already:

-At least 7 route functions, what I've come up with (feel free to change if necessary): (Index, View_exercises, Add_exercise, Remove_exercise, Progress_graph, Clear_graph, Settings_page)

-At least 5 Pages, what I've come up with (feel free to change if necessary): Main Menu, Edit Exercises, Log Exercises, View Progression, Settings

-State Dataclass with at least 4 fields, what I've come up with (change if needed): Exercises: list[Exercise], History: list[int], Streak: int (how many workouts user has done, maybe displayed at top of page), Filter: str

-a Second dataclass, Exercises with fields: Name: str, Muscle_group: str, Sets: int, Reps: int, Weight: int

-At least 3 user inputs -> WeightLifted -> textbox, Sets -> Selectbox, Muscle Group -> SelectBox (in edit exercises), Website Theme: checkbox (settings menu, dark/light theme)

-At least 3 If Statements -> check if the textbox for exercise name is blank when a user creates a new exercise or edits a previous one ; Max weight -> this checks if your current logged weight is greater than your highest ever logged weight and displays that as your max weight when you select that exercise. ; Max weight -> this checks if your current logged weight is greater than your highest ever logged weight and displays that as your max weight when you select that exercise. (Add more if necessary or change if deemed unnecessary)

-At least 1 loop -> iterate over the history list in the State dataclass to grab weights for one particular exercise to be graphed (change if not needed or more if needed)

-Site cannot have global variables but global constants are okay

-Must have thorough code documentation

Again, make changes if needed because my outline may not be 100% sufficient.

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